

RAG Interview Questions and Answers

Q1. [Beginner] What is RAG?

Answer: Retrieval Augmented Generation.

Q1. [Beginner] What is Retrieval-Augmented Generation (RAG)?

Answer: What is Retrieval-Augmented Generation (RAG)? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q2. [Beginner] Why is RAG needed?

Answer: Why is RAG needed? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q3. [Beginner] What are the main components of a RAG pipeline?

Answer: What are the main components of a RAG pipeline? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q4. [Beginner] What is a vector database?

Answer: What is a vector database? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q5. [Beginner] What is an embedding?

Answer: What is an embedding? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q6. [Beginner] Why are embeddings important in RAG?

Answer: Why are embeddings important in RAG? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q7. [Beginner] What is document chunking?

Answer: What is document chunking? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q8. [Beginner] Why is chunking necessary?

Answer: Why is chunking necessary? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q9. [Beginner] What is semantic search?

Answer: What is semantic search? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q10. [Beginner] What is similarity search?

Answer: What is similarity search? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q11. [Beginner] What is Top-K retrieval?

Answer: What is Top-K retrieval? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q12. [Beginner] What is cosine similarity?

Answer: What is cosine similarity? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q13. [Beginner] What is the role of the retriever?

Answer: What is the role of the retriever? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q14. [Beginner] What is the role of the generator?

Answer: What is the role of the generator? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q15. [Beginner] How does RAG reduce hallucinations?

Answer: How does RAG reduce hallucinations? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q16. [Beginner] What is a knowledge base in RAG?

Answer: What is a knowledge base in RAG? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q17. [Beginner] What types of documents can be used in RAG?

Answer: What types of documents can be used in RAG? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q18. [Beginner] What is metadata in a vector database?

Answer: What is metadata in a vector database? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q19. [Beginner] What is query embedding?

Answer: What is query embedding? can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q20. [Beginner] Explain the end-to-end flow of a RAG system.

Answer: Explain the end-to-end flow of a RAG system. can be explained as a core concept in RAG systems. It helps retrieve relevant information and improve the quality and factual grounding of LLM responses.

Q21. [Intermediate] What is fixed-size chunking?

Answer: What is fixed-size chunking? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q22. [Intermediate] What is recursive chunking?

Answer: What is recursive chunking? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q23. [Intermediate] What is semantic chunking?

Answer: What is semantic chunking? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q24. [Intermediate] What is sliding-window chunking?

Answer: What is sliding-window chunking? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q25. [Intermediate] Why is chunk overlap important?

Answer: Why is chunk overlap important? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q26. [Intermediate] What is dense retrieval?

Answer: What is dense retrieval? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q27. [Intermediate] What is sparse retrieval?

Answer: What is sparse retrieval? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q28. [Intermediate] How does BM25 work?

Answer: How does BM25 work? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q29. [Intermediate] What is hybrid search?

Answer: What is hybrid search? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q30. [Intermediate] Compare BM25 and vector search.

Answer: Compare BM25 and vector search. is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q31. [Intermediate] What is metadata filtering?

Answer: What is metadata filtering? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q32. [Intermediate] What is query expansion?

Answer: What is query expansion? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q33. [Intermediate] What is multi-query retrieval?

Answer: What is multi-query retrieval? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q34. [Intermediate] What is a re-ranker?

Answer: What is a re-ranker? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q35. [Intermediate] How do cross-encoders improve retrieval?

Answer: How do cross-encoders improve retrieval? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q36. [Intermediate] What is contextual compression?

Answer: What is contextual compression? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q37. [Intermediate] What is parent-child retrieval?

Answer: What is parent-child retrieval? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q38. [Intermediate] What is a self-query retriever?

Answer: What is a self-query retriever? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q39. [Intermediate] How do you evaluate retrieval quality?

Answer: How do you evaluate retrieval quality? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q40. [Intermediate] What retrieval metrics are commonly used in RAG?

Answer: What retrieval metrics are commonly used in RAG? is used in modern RAG systems to improve retrieval effectiveness and provide higher-quality context for downstream generation.

Q41. [Advanced] What is Context Precision in RAG evaluation?

Answer: What is Context Precision in RAG evaluation? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q42. [Advanced] What is Context Recall in RAG evaluation?

Answer: What is Context Recall in RAG evaluation? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q43. [Advanced] What is Faithfulness in RAGAS?

Answer: What is Faithfulness in RAGAS? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q44. [Advanced] What is Answer Relevancy?

Answer: What is Answer Relevancy? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q45. [Advanced] How does RAGAS evaluate RAG systems?

Answer: How does RAGAS evaluate RAG systems? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q46. [Advanced] How does DeepEval evaluate RAG applications?

Answer: How does DeepEval evaluate RAG applications? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q47. [Advanced] What is Multi-Vector Retrieval?

Answer: What is Multi-Vector Retrieval? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q48. [Advanced] What is ColBERT retrieval?

Answer: What is ColBERT retrieval? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q49. [Advanced] What is Graph RAG?

Answer: What is Graph RAG? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q50. [Advanced] How do knowledge graphs enhance RAG?

Answer: How do knowledge graphs enhance RAG? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q51. [Advanced] What is Agentic RAG?

Answer: What is Agentic RAG? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q52. [Advanced] What is Corrective RAG (CRAG)?

Answer: What is Corrective RAG (CRAG)? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q53. [Advanced] What is Adaptive RAG?

Answer: What is Adaptive RAG? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q54. [Advanced] What is Self-RAG?

Answer: What is Self-RAG? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q55. [Advanced] What is RAPTOR and how does it improve retrieval?

Answer: What is RAPTOR and how does it improve retrieval? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q56. [Advanced] What is multi-hop retrieval?

Answer: What is multi-hop retrieval? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q57. [Advanced] When should long-context models replace RAG?

Answer: When should long-context models replace RAG? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q58. [Advanced] How can caching improve RAG performance?

Answer: How can caching improve RAG performance? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q59. [Advanced] What are common failure modes in RAG systems?

Answer: What are common failure modes in RAG systems? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q60. [Advanced] How do you build a production-grade RAG architecture?

Answer: How do you build a production-grade RAG architecture? is an advanced RAG concept used to improve retrieval quality, evaluation accuracy, scalability, or overall system performance in production environments.

Q61. [Expert] Design a RAG system for 100 million documents.

Answer: Design a RAG system for 100 million documents. should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q62. [Expert] How would you optimize retrieval latency at scale?

Answer: How would you optimize retrieval latency at scale? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q63. [Expert] How do you select an embedding model for enterprise RAG?

Answer: How do you select an embedding model for enterprise RAG? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q64. [Expert] Design a hybrid retrieval architecture for legal documents.

Answer: Design a hybrid retrieval architecture for legal documents. should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q65. [Expert] How would you implement distributed vector search?

Answer: How would you implement distributed vector search? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q66. [Expert] How would you build a multi-tenant RAG platform?

Answer: How would you build a multi-tenant RAG platform? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q67. [Expert] How would you design real-time ingestion for RAG?

Answer: How would you design real-time ingestion for RAG? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q68. [Expert] How would you optimize RAG infrastructure costs?

Answer: How would you optimize RAG infrastructure costs? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q69. [Expert] How would you debug poor retrieval quality in production?

Answer: How would you debug poor retrieval quality in production? should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q70. [Expert] Design an enterprise-grade RAG platform on GCP.

Answer: Design an enterprise-grade RAG platform on GCP. should be approached by balancing retrieval quality, scalability, latency, reliability, cost, and maintainability. The architecture should support efficient retrieval while ensuring accurate and grounded responses.

Q71. [Scenario] Your RAG chatbot is returning irrelevant documents for most user queries. How would you troubleshoot and fix the issue?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q72. [Scenario] Users report that the chatbot frequently hallucinates despite using RAG. What steps would you take?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q73. [Scenario] A document search that previously took 2 seconds now takes 15 seconds. How would you investigate?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q74. [Scenario] The company has 10 million new documents to ingest daily. How would you redesign the ingestion pipeline?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q75. [Scenario] Different customers should only access their own documents in a multi-tenant RAG platform. How would you implement this?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q76. [Scenario] The retrieval system works well for simple questions but fails for multi-hop questions. What improvements would you make?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q77. [Scenario] Your vector database costs have doubled in three months. How would you optimize the system?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q78. [Scenario] The business wants real-time updates from transactional databases reflected in the RAG system within minutes. How would you design it?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q79. [Scenario] Evaluation scores are high but users still complain about answer quality. How would you investigate the gap?

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.

Q80. [Scenario] Design a complete enterprise RAG solution on GCP for a global organization with strict security requirements.

Answer: I would first identify where the failure occurs in the pipeline—ingestion, embeddings, retrieval, ranking, context construction, or generation. Then I would collect metrics, validate assumptions, implement fixes, and measure improvements through evaluation frameworks and user feedback.